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**ROBUST CONTROL OF LINEAR SYSTEMS WITH
SWITCHED ACTUATORS SUBJECTED TO
DWEELL-TIME CONSTRAINTS**

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ROBUST CONTROL OF LINEAR SYSTEMS WITH SWITCHED ACTUATORS SUBJECTED TO DWELL-TIME CONSTRAINTS

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ITA

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*"I am and always will be the optimist.
The hoper of far-flung hopes, and the dreamer of improbable dreams."*
— THE ELEVENTH DOCTOR

Resumo

Atuadores chaveados são difíceis de lidar por conta de algumas restrições características de sua natureza, porém eles são eficientes em inúmeras aplicações da engenharia, sendo aplicado, por exemplo, em sistemas de controle de atitude de veículos espaciais. Assim, saber utilizar desse tipo de atuador se faz importante e necessário. O principal problema a ser resolvido nessa tese é o projeto de um controlador que garanta a robustez de sistema com atuadores chaveados submetidos a restrições temporais que limitam a janela de resposta de seus comandos. Nessa abordagem, é feita uma busca de trajetória alvo de ciclo limite e, então, é calculado o erro dos estados e aplicada a técnica de controle preditivo para correção do erro e levar o sistema à trajetória cíclica alvo.

Abstract

Switched actuators are difficult to handle because of some characteristic constraints of their nature. but they are efficient in numerous engineering applications, being applied, for example, in attitude control systems for space vehicles. Thus, knowing how to use this type of actuator is important and necessary. The main problem to be solved in this thesis is the design of a controller that guarantees the robustness of a system with switched actuators subjected to time constraints that limit the response window of their commands. In this approach, a limit cycle target trajectory search is performed and then the error of the states compared to the target trajectory is calculated. Then a predictive control technique is applied to correct the error and get the system to the cyclic target trajectory

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List of Abbreviations and Acronyms

CTq	computed torque
DC	direct current
EAR	Equação Algébrica de Riccati
GDL	graus de liberdade
ISR	interrupção de serviço e rotina
LMI	linear matrices inequalities
MIMO	multiple input multiple output
PD	proporcional derivativo
PID	proporcional integrativo derivativo
PTP	point to point
UARMII	Underactuated Robot Manipulator II
VSC	variable structure control

List of Symbols

a	Distância
$\mathbf{a}$	Vetor de distâncias
$\mathbf{e}_j$	Vetor unitário de dimensão n e com o j -ésimo componente igual a 1
$\mathbf{K}$	Matriz de rigidez
m_1	Massa do cumpim
δ_{k-k_f}	Delta de Kronecker no instante k_f

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1 Introduction

1.1 Motivation

Switched affine systems, $\dot{x}(t) = A_{\sigma(t)}x(t) + B_{\sigma(t)}$, can describe several problems in modern engineering over many different areas. Most of the proposed controllers solve this by driving the system states to a region close to the target state and, for a affine switched system, this means that the system will find stability in a periodic trajectory near the target but never reaching the exactly desired states. Very often, the form of this periodic trajectory relates to the performance of the system, but this is something that is not addressed with the simpler approaches of controllers.

referencias do problema na literatura

Some research studies investigated the solution of the problem addressing the target cyclic trajectory (BENMILLOUD *et al.*, 2019). In this work, the author proposes a control design that transitates between successive discrete modes inside the limit cycle.

(VIEIRA, 2013) solved the problem for a particular class of affine systems. Here the controller drove the system to a target state using MILP (Mixed-Integer Linear Programming) and the cyclic trajectory was a consequence of the states as the system approached the target.

$$\dot{x}(t) = A_c x(t) + B_c u(t)$$

patino, entre marcio e mateus. Falar sobre o trabalho do patino aqui

(MARCOLINO, 2018) proposed a solution with a controller law that drives the system to a target cyclic trajectory. In this work, the problem was divided in two parts: first there is a stage that uses optimization to find a targeted ideal cyclic trajectory, then a second part that uses that target trajectory as input and applies perturbation on the instants of switching of the control signal using predictive control.

1.2 Contributions of the present work

This present work proposes to expand the solution from (MARCOLINO, 2018) and provide a broader solution that can be used for systems that have their dynamics changing over time.

Examples of this kind of system is: switched circuits that behaves with a different dynamics after each switching atuation.

1.3 Outline of the remaining chapters

The remaining of this text is organized as follows. Chapter 2 describes the problem of finding a target periodic trajectory for the given switched affine system. Chapter 3 proposes then a predictive control law that drives the system to the target cyclic trajectory. Finally, chapter 4 brings some remarks and considerations of the work.

2 Determination of Target Periodic Trajectories

Devemos encontrar uma trajetória cíclica que respeite todos as restrições do sistema e minimize a distancia em torno do valor de referencia?

Função custo:

$$J(\tau^\infty, \Omega^\infty, s^\infty) = \min_{\tau, \Omega, s} \int_0^{T_p} [x(t) - \bar{x}^\infty]^T Q [x(t) - \bar{x}^\infty] dt \quad (2.1)$$

As entradas para a otimização são:

- Lista de modos de operação
- Lista de matriz A associada ao modo de operação
- Lista de matriz B associada ao modo de operação
- Matriz C
- Matriz D
- Lista de valores de controle em cada modo
- Valor de referência dos estados (xref)
- Tempo máximo de trajetória cíclica (Tpmax)
- Número máximo de estados
- Peso dos estados na função custo (Q)

Em que $Q = Q^T > 0$ e T_p é o período do ciclo.

A condição inicial é determinada de forma que o último ponto é coincidente com o primeiro ponto (dada a sequência de modos e instantes de chaveamento)

$$x(0) = (I - F_N F_{N-1} \dots F_1)^{-1} c \quad (2.2)$$

$$c = F_N F_{N-1} \dots F_2 G_1 u_1 + F_N F_{N-1} \dots F_3 G_2 u_2 + \dots + F_N G_{N-1} + G_N u_N \quad (2.3)$$

O resultado da otimizacao é τ^∞ e x_0

Exemplos:

Integrador (Matheus): Nao há otimizacao aqui, usamos a mesma trajetoria do Matheus para que tivessemos uma comparacao entre os resultados das simulacoes.

Patino, aqui estou usando um otimizador para encontrar a trajetoria de referencia.

Resultados:

Descrever resultados trajetorias Patino 1 e 2

TODO: FAZER SIMULACAO COMPLETA DE CADA CASO

2.1 Statement of the problem

2.2 Optimization

description of the optimization for finding the target trajectory

2.3 Calculating x_r

$x_0 \in \mathbb{R}$

$$x(t) = e^{At} x(0) + \int_0^t e^{A(t-\tau)} B u(\tau) d\tau \quad (2.4)$$

Modos:

$$\begin{aligned} [0 - t_1] : \dot{x} &= A_1 x + B_1 u_1 \\ [t_1 - t_2] : \dot{x} &= A_2 x + B_2 u_2 \\ &\vdots \\ [t_{N-1} - t_N] : \dot{x} &= A_N x + B_N u_N \end{aligned} \quad (2.5)$$

$$\begin{aligned}
x(t_1) &= e^{A_1 t_1} x(0) + \int_0^{t_1} e^{A_1(t_1-\tau)} B_1 u_1 d\tau \\
x(t_1) &= e^{A_1 t_1} x(0) + \left[\int_0^{t_1} e^{A_1(t_1-\tau)} B_1 d\tau \right] u_1 \\
x(t_1) &= F_1 x(0) + G_1 u_1
\end{aligned} \tag{2.6}$$

$$x(t_1) = F_1 x(0) + G_1 u_1 \tag{2.7}$$

$$x(t_2) = F_2 x(1) + G_2 u_2 \tag{2.8}$$

$$\vdots$$

$$x(t_N) = F_N x(N-1) + G_N u_N \tag{2.9}$$

$$\begin{aligned}
x(t_N) &= F_N F_{N-1} \dots F_1 x(0) + F_N F_{N-1} \dots F_2 G_1 x(1) + \\
&\quad F_N F_{N-1} \dots F_2 G_2 x(2) + \dots + F_N G_{N-1} x(N-1) + G_N x(N) \\
x(t_N) &= F_N F_{N-1} \dots F_1 x(0) + c
\end{aligned} \tag{2.10}$$

Queremos $x(t_N) = x(0) = ?$

$$\begin{aligned}
x(0) &= F_N F_{N-1} \dots F_1 x(0) + c \\
\underbrace{(I - F_N F_{N-1} \dots F_1)}_{\text{suposta invertível}} x(0) &= c \\
x(0) &= (I - F_N F_{N-1} \dots F_1)^{-1} c
\end{aligned} \tag{2.11}$$

em que

$$\begin{aligned}
c &= F_N F_{N-1} \dots F_2 G_1 x(1) + \\
&\quad F_N F_{N-1} \dots F_2 G_2 x(2) + \dots + F_N G_{N-1} x(N-1) + G_N x(N)
\end{aligned} \tag{2.12}$$

2.4 Application Examples

2.4.1 Variant Dynamic Circuit 1

2.4.2 Variant Dynamic Circuit 2

3 Predictive Control for Stabilization of Periodic Trajectories

3.1 Statement of the problem

considerando a planta de um sistema chaveado afim descrito por:

$$\dot{x}(t) = A_{\sigma}(t)x(t) + B_{\sigma}(t) \quad (3.1)$$

submetido a restrições de tempo de chaveamento.

A ação de controle aplicada no sistema é na forma de perturbacoes no tempo de chaveamento de cada modo de operação, e, para podermos controlar o sistema com controle preditivo, precisamos encontrar uma equacao linear para trabalhar a relação entre os instantes de chaveamento e o estado resultante do sistema ao final de um ciclo.

PROBLEMA DE CONTROLE ÓTIMO

funcao custo

$$J = \sum_{i=1}^N \hat{e}_{k+i|k} Q \hat{e}_{k+i|k}^T + \rho \hat{\delta t}_{k+i|k}^2 \quad (3.2)$$

em que $Q = Q^T > 0$, $\rho > 0$

EQUACAO DE PRECIÇÃO

$$\hat{x} = G \hat{\delta t} + f \quad (3.3)$$

$$\hat{\mathbf{x}} = \begin{bmatrix} \hat{\mathbf{x}}(t_{r_{k+1}} + \hat{\delta t}_{k+1|k}) \\ \hat{\mathbf{x}}(t_{r_{k+2}} + \hat{\delta t}_{k+1|k} + \hat{\delta t}_{k+2|k}) \\ \vdots \\ \hat{\mathbf{x}}(t_{r_{k+N}} + \hat{\delta t}_{k+1|k} + \dots + \hat{\delta t}_{k+N|k}) \end{bmatrix} \quad (3.4)$$

$$\hat{\delta \mathbf{t}} = \begin{bmatrix} \hat{\delta t}_{k+1|k} \\ \hat{\delta t}_{k+2|k} \\ \vdots \\ \hat{\delta t}_{k+N|k} \end{bmatrix} \quad (3.5)$$

$$\mathbf{f} = \begin{bmatrix} \mathbf{f}_{k+1|k} \\ \mathbf{f}_{k+2|k} \\ \vdots \\ \mathbf{f}_{k+N|k} \end{bmatrix} \quad (3.6)$$

$$\mathbf{G} = \begin{bmatrix} G_{1,1} & 0 & 0 & \dots & 0 \\ G_{2,1} & G_{2,2} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ G_{N,1} & G_{N,2} & G_{N,3} & \dots & G_{N,N} \end{bmatrix} \quad (3.7)$$

Now, let $\mathbf{x}^r$ be the vector of target values defined as

$$\mathbf{x}^r = \begin{bmatrix} x_r(t_{r_{k+1}}) \\ x_r(t_{r_{k+2}}) \\ \vdots \\ x_r(t_{r_{k+N}}) \end{bmatrix} \quad (3.8)$$

In view of [completar texto aqui](#), the cost function is rewritten as

$$J = (\hat{\mathbf{x}} - \mathbf{x}^r)^T \mathbf{Q} (\hat{\mathbf{x}} - \mathbf{x}^r) + \rho \hat{\delta \mathbf{t}}^T \hat{\delta \mathbf{t}} \quad (3.9)$$

where $\mathbf{Q} = \mathbf{Q}^T > 0$ is an $nN \times nN$ block-diagonal matrix defined as $\mathbf{Q} = \text{diag}(Q, Q, \dots, Q)$.

CONSTRAINTS

As restricoes do controlador devem descrever os limites de operacao da dinamica do modelo e ainda encontrar uma soluç o que leve o sistema para mais proximo da referencia.

ESCREVER EQUACOES DE RESTRICAO

descrever que a trajetoria deve ser admissivel e forma com que o sistema calcula o

controle tambem deve ser admissivel

fazer um rapido comentario sobre a forma que essa restricao é trabalhada no capitulo 2

FEASIBILITY REGION

nao aplicavel ????

como calcular ??? restricoes do problem de prog quad

sobre o controle

variacoes do instante de chaveadmento

mas envolvem o estado inicial do ciclo

sea dependencia das restricoes com respeito ao estao inicial for linear

entao podemos encarar o x_0 como uma variavel junto com as variaveis de controle

com isso as restricoes descrevem um polietro aumenta do de controle e estado inicial

geometria computacional mpttoolbox

projeta a geometria no subespaco

pode haver melhoria com o ponto final livre

interessante para avaliar o desempenho do controlador na regio de factibilidade

APPLICATION EXAMPLES

- Integrador duplo(matheus)
- Patino 1
- Patino 2

CONCLUSION

mostrar o modelo em malha aberta indo para a sua trajetoria ciclica estavel junto com o modelo com controle ativo mostrando a convergencia mais rapida

3.2 Linearization procedure

The solution fo the state equation (??) at time t_N , starting from the initial condition $x(t_0)$, is of the form

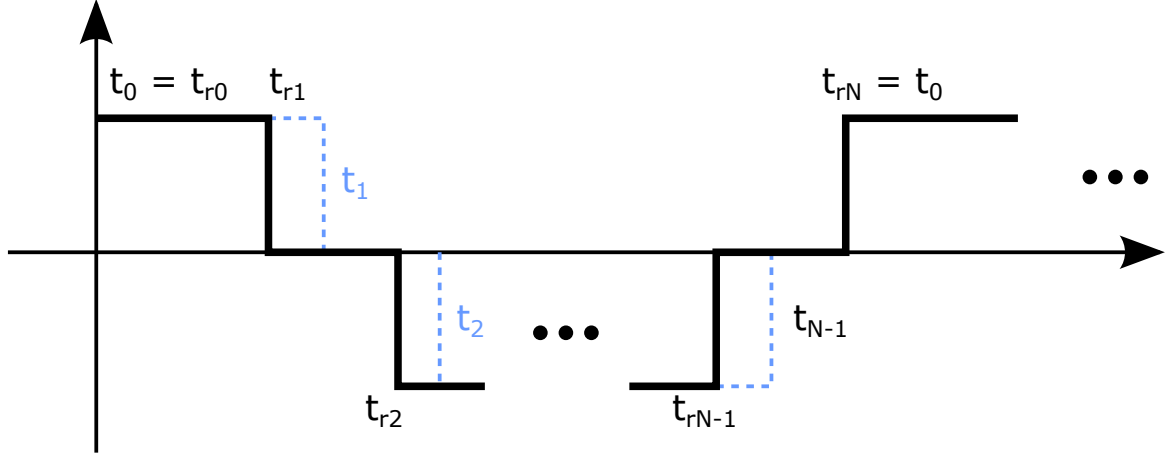


FIGURE 3.1 – Instantes de chaveamento

$$x(t_1) = e^{A_0(t_1-t_0)}x(t_0) + \int_{t_0}^{t_1} e^{A_0(t_1-\tau)} B_0 d\tau \quad (3.10)$$

$$x(t_{r_1}) = e^{A_0(t_{r_1}-t_{r_0})}x(t_{r_0}) + \int_{t_{r_0}}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau \quad (3.11)$$

subtracting 3.10 and 3.11 we get

$$\begin{aligned} x(t_1) - x(t_{r_1}) &= \overbrace{e^{A_0(t_1-t_0)}x(t_0) - e^{A_0(t_{r_1}-t_{r_0})}x(t_{r_0})}^{\square} \\ &+ \underbrace{\int_{t_0}^{t_1} e^{A_0(t_1-\tau)} B_0 d\tau - \int_{t_{r_0}}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau}_{\bigcirc} \end{aligned} \quad (3.12)$$

expanding $\square$

$$\begin{aligned} \square &= e^{A_0(t_1-t_0)}x(t_0) - e^{A_0(t_{r_1}-t_{r_0})}x(t_{r_0}) \\ &= e^{A_0(t_{r_1}+\delta t_1-t_0)}x(t_0) - e^{A_0(t_{r_1}-t_{r_0})}x(t_{r_0}) \\ &= e^{A_0(t_{r_1}-t_0)}e^{\delta t_1}x(t_0) - e^{A_0(t_{r_1}-t_{r_0})}x(t_{r_0}) \\ &= F_{r_0}e^{\delta t_1}x(t_0) - F_{r_0}x(t_{r_0}) \\ &= F_{r_0}(I + A_0\delta t_1)x(t_0) - F_{r_0}x(t_{r_0}) \\ &= F_{r_0}[x(t_0) - x_r(t_0)] + F_{r_0}A_0\delta t_1 \end{aligned} \quad (3.13)$$

expanding $\bigcirc$

$$\begin{aligned}
 \bigcirc &= \int_{t_0}^{t_1} e^{A_0(t_1-\tau)} B_0 d\tau - \int_{t_{r_0}}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau \\
 &= \int_{t_0}^{t_{r_1}+\delta t_1} e^{A_0(t_{r_1}+\delta t_1-\tau)} B_0 d\tau - \int_{t_0}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau \\
 &= e^{A_0\delta t_1} \left[\int_{t_0}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau + \int_{t_{r_1}}^{t_{r_1}+\delta t_1} e^{A_0(t_{r_1}-\tau)} B_0 d\tau \right] - \int_{t_0}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau \\
 &= e^{A_0\delta t_1} [G_{r_0} + B_0\delta t_1] - G_{r_0} \\
 &= (I + A_0\delta t_1)[G_{r_0} + B_0\delta t_1] - G_{r_0} \\
 &= \cancel{G_{r_0}} + B_0\delta t_1 + A_0G_{r_0}\delta t_1 + \cancel{A_0B_0\delta t_1\delta t_1} - \cancel{G_{r_0}} \\
 &= A_0G_{r_0}\delta t_1 + B_0\delta t_1
 \end{aligned} \tag{3.14}$$

and replacing (3.13) and (3.14) into equation 3.12 we get to a equation that computes $x(t_1) - x_r(t_{r_1})$ from $x(t_0) - x_r(t_{r_0})$

$$x(t_1) - x_r(t_{r_1}) = F_{r_0}[x(t_0) - x_r(t_0)] + [F_{r_0}A_0x(t_0) + A_0G_{r_0} + B_0]\delta t_1 \tag{3.15}$$

Repeating the proceduure for $x(t_2)$ we have

$$x(t_2) = e^{A_1(t_2-t_1)}x(t_1) + \int_{t_1}^{t_2} e^{A_1(t_2-\tau)} B_1 d\tau \tag{3.16}$$

$$x(t_{r_2}) = e^{A_1(t_{r_2}-t_{r_1})}x(t_{r_1}) + \int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)} B_1 d\tau \tag{3.17}$$

subtracting 3.16 and 3.17

$$\begin{aligned}
 x(t_2) - x(t_{r_2}) &= \overbrace{e^{A_1(t_2-t_1)}x(t_1) - e^{A_1(t_{r_2}-t_{r_1})}x(t_{r_1})}^{\square} \\
 &\quad + \underbrace{\int_{t_1}^{t_2} e^{A_1(t_2-\tau)} B_1 d\tau - \int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)} B_1 d\tau}_{\bigcirc}
 \end{aligned} \tag{3.18}$$

expanding $\square$

$$\begin{aligned}
 \square &= e^{A_1(t_{r_2}+\delta t_2-t_{r_1}-\delta t_1)}x(t_1) - e^{A_1(t_{r_2}-t_{r_1})}x_r(t_{r_1}) \\
 &= e^{A_1(t_{r_2}-t_{r_1})}e^{A_1(\delta t_2-\delta t_1)}x(t_1) - e^{A_1(t_{r_2}-t_{r_1})}x_r(t_{r_1}) \\
 &= F_{r_1}[I + A_1(\delta t_2 - \delta t_1)]x(t_1) - F_{r_1}x_r(t_{r_1}) \\
 &= F_{r_1}[x(t_1) - x_r(t_{r_1})] + F_{r_1}A_1x(t_1)\delta t_2 - F_{r_1}A_1x(t_1)\delta t_1
 \end{aligned} \tag{3.19}$$

expanding $\bigcirc$

$$\begin{aligned}
 \bigcirc &= \int_{t_1}^{t_2} e^{A_1(t_2-\tau)}B_1d\tau - \int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)}B_1d\tau \\
 &= \int_{t_{r_1}+\delta t_1}^{t_{r_2}+\delta t_2} e^{A_1(t_{r_2}+\delta t_2-\tau)}B_1d\tau - \int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)}B_1d\tau \\
 &= e^{A_1\delta t_2} \left[\int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)}B_1d\tau - \int_{t_{r_1}}^{t_{r_1}+\delta t_1} e^{A_1(t_{r_2}-\tau)}B_1d\tau + \int_{t_{r_2}}^{t_{r_2}+\delta t_2} e^{A_1(t_{r_2}-\tau)}B_1d\tau \right] \\
 &\quad - \int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)}B_1d\tau \\
 &= (I + A_1\delta t_2) \left[G_{r_1} - \int_{t_{r_1}}^{t_{r_1}+\delta t_1} e^{A_1(t_{r_2}-\tau)}B_1d\tau + \int_{t_{r_2}}^{t_{r_2}+\delta t_2} e^{A_1(t_{r_2}-\tau)}B_1d\tau \right] - G_{r_1} \\
 &= (I + A_1\delta t_2) \left[G_{r_1} - e^{A_1(t_{r_2}-t_{r_1})}B_1\delta t_1 + B_1\delta t_2 \right] - G_{r_1} \\
 &= (I + A_1\delta t_2) \left[G_{r_1} - F_{r_1}B_1\delta t_1 + B_1\delta t_2 \right] - G_{r_1} \\
 &= \cancel{G_{r_1}} - F_{r_1}B_1\delta t_1 + B_1\delta t_2 + A_1G_{r_1}\delta t_2 - \cancel{G_{r_1}} \\
 &= A_1G_{r_1}\delta t_2 + B_1\delta t_2 - F_{r_1}B_1\delta t_1
 \end{aligned} \tag{3.20}$$

replacing (3.19) and (3.20) into equation (3.18)

$$\begin{aligned}
 x(t_2) - x_r(t_{r_2}) &= F_{r_1}[x(t_1) - x_{r_1}(t_{r_1})] \\
 &\quad - F_{r_1}[A_1x(t_1) + B_1]\delta t_1 \\
 &\quad + [F_{r_1}A_1 \underbrace{x(t_1)}_{\triangle} + A_1G_{r_1} + B_1]\delta t_2
 \end{aligned} \tag{3.21}$$

expanding $\triangle$ from 3.15

$$\triangle = x(t_1) = x_r(t_{r_1}) + F_{r_0}[x(t_0) - x_r(t_0)] + [F_{r_0}A_0x(t_0) + A_0G_{r_0} + B_0]\delta t_1$$

note that when $\triangle$ is replaced in eq. 3.21 it will produce a product with δt_1 and δt_2 like

$$x(t_1)\delta t_1 = x_r(t_{r_1})\delta t_1 + F_{r_0} \overbrace{[x(t_0) - x_r(t_0)]\delta t_1}^{\square} + [F_{r_0}A_0x(t_0) + A_0G_{r_0} + B_0] \overbrace{\delta t_1\delta t_1}^{\circ}$$

both $\square$ and $\circ$ are products of differences that produces a smaller number and they will computed as ≈ 0

$$x(t_1)\delta t_1 \approx x_r(t_{r_1})\delta t_1 \quad (3.22)$$

replacing equation 3.15 and 3.22 in 3.21

$$\begin{aligned} x(t_2) - x_r(t_{r_2}) &= F_{r_1} \{ F_{r_0} [x(t_0) - x_r(t_0) + [F_{r_0}A_0x(t_0) + A_0G_{r_0} + B_0]\delta t_1] \\ &\quad - F_{r_1} [A_1x_r(t_1) + B_1]\delta t_1 \\ &\quad + [F_{r_1}A_1x_r(t_1) + A_1G_{r_1} + B_1]\delta t_2 \} \\ x(t_2) - x_r(t_{r_2}) &= F_{r_1}F_{r_0}[x(t_0) - x_r(t_0)] \\ &\quad + F_{r_1}[F_{r_0}A_0x(t_0) + A_0G_{r_0} - A_1x_r(t_{r_1}) + B_0 - B_1]\delta t_1 \\ &\quad + [F_{r_1}A_1x_r(t_{r_1}) + A_1G_{r_1} + B_1]\delta t_2 \end{aligned} \quad (3.23)$$

Now, repeating the procedure for $x(t_3)$ we have

$$x(t_3) = e^{A_2(t_3-t_2)}x(t_2) + \int_{t_2}^{t_3} e^{A_2(t_3-\tau)}B_2d\tau \quad (3.24)$$

$$x(t_{r_3}) = e^{A_2(t_{r_3}-t_{r_2})}x(t_{r_2}) + \int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3}-\tau)}B_2d\tau \quad (3.25)$$

subtracting 3.24 and 3.25

$$\begin{aligned} x(t_3) - x(t_{r_3}) &= \overbrace{e^{A_2(t_3-t_2)}x(t_2) - e^{A_2(t_{r_3}-t_{r_2})}x(t_{r_2})}^{\square} \\ &\quad + \underbrace{\int_{t_2}^{t_3} e^{A_2(t_3-\tau)}B_2d\tau - \int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3}-\tau)}B_2d\tau}_{\circ} \end{aligned} \quad (3.26)$$

expanding $\square$

$$\begin{aligned}
 \square &= e^{A_2(t_{r_3}+\delta t_3-t_{r_2}-\delta t_2)}x(t_2) - e^{A_2(t_{r_3}-t_{r_2})}x_r(t_{r_2}) \\
 &= e^{A_2(t_{r_3}-t_{r_2})}e^{A_2(\delta t_3-\delta t_2)}x(t_2) - e^{A_2(t_{r_3}-t_{r_2})}x_r(t_{r_2}) \\
 &= F_{r_2}[I + A_2(\delta t_3 - \delta t_2)]x(t_2) - F_{r_2}x_r(t_{r_2}) \\
 &= F_{r_2}[x(t_2) - x_r(t_{r_2})] + F_{r_2}A_2x(t_2)\delta t_3 - F_{r_2}A_2x(t_2)\delta t_2
 \end{aligned} \tag{3.27}$$

expanding $\bigcirc$

$$\begin{aligned}
 \bigcirc &= \int_{t_2}^{t_3} e^{A_2(t_3-\tau)}B_2d\tau - \int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3}-\tau)}B_2d\tau \\
 &= \int_{t_{r_2}+\delta t_2}^{t_{r_3}+\delta t_3} e^{A_2(t_{r_3}+\delta t_3-\tau)}B_2d\tau - \int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3}-\tau)}B_2d\tau \\
 &= e^{A_2\delta t_3} \left[\int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3}-\tau)}B_2d\tau - \int_{t_{r_2}}^{t_{r_2}+\delta t_2} e^{A_2(t_{r_3}-\tau)}B_2d\tau + \int_{t_{r_3}}^{t_{r_3}+\delta t_3} e^{A_2(t_{r_3}-\tau)}B_2d\tau \right] \\
 &\quad - \int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3}-\tau)}B_2d\tau \\
 &= (I + A_2\delta t_3) \left[G_{r_2} - \int_{t_{r_2}}^{t_{r_2}+\delta t_2} e^{A_2(t_{r_3}-\tau)}B_2d\tau + \int_{t_{r_3}}^{t_{r_3}+\delta t_3} e^{A_2(t_{r_3}-\tau)}B_2d\tau \right] - G_{r_2} \\
 &= (I + A_2\delta t_3) \left[G_{r_2} - e^{A_2(t_{r_3}-t_{r_2})}B_2\delta t_2 + B_2\delta t_3 \right] - G_{r_2} \\
 &= (I + A_2\delta t_3) \left[G_{r_2} - F_{r_2}B_2\delta t_2 + B_2\delta t_3 \right] - G_{r_2} \\
 &= \cancel{G_{r_2}} - F_{r_2}B_2\delta t_2 + B_2\delta t_3 + A_2G_{r_2}\delta t_3 - \cancel{G_{r_2}} \\
 &= A_2G_{r_2}\delta t_3 + B_2\delta t_3 - F_{r_2}B_2\delta t_2
 \end{aligned} \tag{3.28}$$

replacing (3.27) in (3.28) back in eq. (3.26)

$$\begin{aligned}
 x(t_3) - x_r(t_{r_3}) &= F_{r_2}[x(t_2) - x_{r_2}(t_{r_2})] \\
 &\quad - F_{r_2}[A_2x(t_2) + B_2]\delta t_2 \\
 &\quad + [F_{r_2}A_2x(t_2) + A_2G_{r_2} + B_2]\delta t_3
 \end{aligned}$$

$$\begin{aligned}
 x(t_3) - x_r(t_{r_3}) = & F_{r_2} \{ F_{r_1} F_{r_0} [x(t_0) - x_r(t_0)] \\
 & + F_{r_1} [F_{r_0} A_0 x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1] \delta t_1 \\
 & + [F_{r_1} A_1 x_r(t_1) + A_1 G_{r_1} + B_1] \delta t_2 \} \\
 & - F_{r_2} [A_2 x_r(t_2) + B_2] \delta t_2 \\
 & + [F_{r_2} A_2 x_r(t_2) + A_2 G_{r_2} + B_2] \delta t_3
 \end{aligned}$$

$$\begin{aligned}
 x(t_3) - x_r(t_{r_3}) = & F_{r_2} F_{r_1} F_{r_0} [x(t_0) - x_r(t_0)] \\
 & + F_{r_2} F_{r_1} [F_{r_0} A_0 x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1] \delta t_1 \\
 & + F_{r_2} [F_{r_1} A_1 x_r(t_1) + A_1 G_{r_1} - A_2 x_r(t_{r_2}) + B_1 - B_2] \delta t_2 \\
 & + [F_{r_2} A_2 x_r(t_2) + A_2 G_{r_2} + B_2] \delta t_3
 \end{aligned} \tag{3.29}$$

Continuing on expanding the propagation of the system further more we can deduce the equation for $x(t_i)$ as

$$\begin{aligned}
 x(t_i) - x_r(t_{r_i}) = & F_{r_{i-1}} F_{r_{i-2}} \dots F_{r_0} [x(t_0) - x_r(t_0)] \\
 & + F_{r_{i-1}} F_{r_{i-2}} \dots F_{r_1} [F_{r_0} A_0 x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1] \delta t_1 \\
 & + F_{r_{i-1}} F_{r_{i-2}} \dots F_{r_2} [F_{r_1} A_1 x_r(t_1) + A_1 G_{r_1} - A_2 x_r(t_{r_2}) + B_1 - B_2] \delta t_2 \\
 & \vdots \\
 & + F_{r_{i-1}} [F_{r_{i-2}} A_{i-2} x_r(t_{i-2}) + A_{i-2} G_{r_{i-2}} - A_{i-1} x_r(t_{r_{i-1}}) + B_{i-2} - B_{i-1}] \delta t_{i-1} \\
 & + [F_{r_{i-1}} A_{i-1} x_r(t_{i-1}) + A_{i-1} G_{r_{i-1}} + B_{i-1}] \delta t_i
 \end{aligned} \tag{3.30}$$

and then, for $x(t_N)$ we have

$$\begin{aligned}
 x(t_N) - x_r(t_{r_N}) = & F_{r_{N-1}} F_{r_{N-2}} \cdots F_{r_0} [x(t_0) - x_r(t_0)] \\
 & + F_{r_{N-1}} F_{r_{N-2}} \cdots F_{r_1} \overbrace{[F_{r_0} A_0 x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1]}^{\square} \delta t_1 \\
 & + F_{r_{N-1}} F_{r_{N-2}} \cdots F_{r_2} [F_{r_1} A_1 x(t_1) + A_1 G_{r_1} - A_2 x_r(t_{r_2}) + B_1 - B_2] \delta t_2 \\
 & \vdots \\
 & + F_{r_{N-1}} [F_{r_{N-2}} A_{N-2} x_r(t_{N-2}) + A_{N-2} G_{r_{N-2}} - A_{N-1} x_r(t_{r_{N-1}}) \\
 & \quad + B_{N-2} - B_{N-1}] \delta t_{N-1} \\
 & + \underbrace{[F_{r_{N-1}} A_{N-1} x_r(t_{N-1}) + A_{N-1} G_{r_{N-1}} + B_{N-1}]}_{\bigcirc} \delta t_N
 \end{aligned} \tag{3.31}$$

$$\begin{aligned}
 \square = & F_{r_0} A_0 x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1 \\
 = & A_0 F_{r_0} x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1 \\
 = & A_0 [x_r(t_{r_1}) - G_{r_0}] + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1 \\
 = & A_0 x_r(t_{r_1}) - \cancel{A_0 G_{r_0}} + \cancel{A_0 G_{r_0}} - A_1 x_r(t_{r_1}) + B_0 - B_1 \\
 = & (A_0 - A_1) x_r(t_{r_1}) + B_0 - B_1
 \end{aligned} \tag{3.32}$$

$$\begin{aligned}
 \bigcirc = & F_{r_{N-1}} A_{N-1} x_r(t_{N-1}) + A_{N-1} G_{r_{N-1}} + B_{N-1} \\
 = & A_{N-1} F_{r_{N-1}} x_r(t_{N-1}) + A_{N-1} G_{r_{N-1}} + B_{N-1} \\
 = & A_{N-1} [x_r(t_{r_N}) - G_{r_{N-1}}] + A_{N-1} G_{r_{N-1}} + B_{N-1} \\
 = & A_{N-1} x_r(t_{r_N}) - \cancel{A_{N-1} G_{r_{N-1}}} + \cancel{A_{N-1} G_{r_{N-1}}} + B_{N-1} \\
 = & A_{N-1} x_r(t_{r_N}) + B_{N-1}
 \end{aligned} \tag{3.33}$$

replacing 3.32 and 3.33 in 3.31

$$\begin{aligned}
 x(t_N) - x_r(t_{r_N}) = & F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_0} [x(t_0) - x_r(t_0)] \\
 & + F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_1} [(A_0 - A_1)x_r(t_{r_1}) + B_0 - B_1] \delta t_1 \\
 & + F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_2} [(A_1 - A_2)x_r(t_{r_2}) + B_1 - B_2] \delta t_2 \\
 & \vdots \\
 & + F_{r_{N-1}} [(A_{N-2} - A_{N-1})x_r(t_{r_{N-1}}) + B_{N-2} - B_{N-1}] \delta t_{N-1} \\
 & + [A_{N-1}x_r(t_{r_N}) + B_{N-1}] \delta t_N
 \end{aligned} \tag{3.34}$$

$$e(t) = x(t) - x_r(t_r) \tag{3.35}$$

$$\Phi = F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_0} \tag{3.36}$$

$$\begin{aligned}
 \Gamma = & \begin{bmatrix} F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_1} [(A_0 - A_1)x_r(t_{r_1}) + B_0 - B_1], \\ F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_2} [(A_1 - A_2)x_r(t_{r_2}) + B_1 - B_2], \\ \vdots \\ F_{r_{N-1}} [(A_{N-2} - A_{N-1})x_r(t_{r_{N-1}}) + B_{N-2} - B_{N-1}] \end{bmatrix}
 \end{aligned} \tag{3.37}$$

$$\delta t = [\delta t_1, \delta t_2, \dots, \delta t_N]^T \tag{3.38}$$

$$e(t_N) = \Phi e(t_0) + \Gamma \delta t \tag{3.39}$$

where x_r is calculated as

$$x_r(t_{r_1}) = F_0 x_r(t_{r_0}) + G_0 \tag{3.40}$$

$$x_r(t_{r_2}) = F_1 x_r(t_{r_1}) + G_1 \tag{3.41}$$

$$\vdots$$

$$x_r(t_{r_N}) = F_{N-1} x_r(t_{r_{N-1}}) + G_{N-1} \tag{3.42}$$

and F_i and G_i are calculated as

$$[F_i, G_i] = c2dm(Ai, Bi, [], [], dti, 'zoh') \tag{3.43}$$

3.3 Optimal control problem

3.3.1 Cost function

3.3.2 Prediction equation

3.3.3 Constraints

3.3.4 Feasibility region

3.4 Application Examples

3.4.1 Integrador duplo

3.4.2 Patino 1

3.4.3 Patino 2

Matrizes do artigo patino exemplo 2:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} -\frac{x_3(t)}{C_1} & \frac{x_3(t)}{C_1} & 0 \\ 0 & -\frac{x_3(t)}{C_2} & \frac{x_3(t)}{C_2} \\ \frac{x_1(t)}{L} & -\frac{x_2(t)-x_1(t)}{L} & \frac{E-x_2(t)}{L} \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -\frac{R}{L}x_3(t) \end{bmatrix} \quad (3.44)$$

$$A = \begin{bmatrix} -\frac{x_3(t)}{C_1} & \frac{x_3(t)}{C_1} & 0 \\ 0 & -\frac{x_3(t)}{C_2} & \frac{x_3(t)}{C_2} \\ \frac{x_1(t)}{L} & \frac{x_2(t)-x_1(t)}{L} & \frac{E-x_2(t)}{L} \end{bmatrix} \quad (3.45)$$

$$\begin{aligned}
 \mathbf{u}_0 &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T \\
 \mathbf{u}_1 &= \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T \\
 \mathbf{u}_2 &= \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T \\
 \mathbf{u}_3 &= \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}^T \\
 \mathbf{u}_4 &= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T \\
 \mathbf{u}_5 &= \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T \\
 \mathbf{u}_6 &= \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}^T \\
 \mathbf{u}_7 &= \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T
 \end{aligned} \tag{3.46}$$

$$A_{u_0} = A\mathbf{u}_0 = \begin{bmatrix} 0 \\ 0 \\ \frac{-Rx_3(t)}{L} \end{bmatrix} \tag{3.47}$$

$$A_{u_1} = A\mathbf{u}_1 = \begin{bmatrix} 0 \\ \frac{x_3(t)}{C_2} \\ \frac{E-x_2(t)-Rx_3(t)}{L} \end{bmatrix} \tag{3.48}$$

$$A_{u_2} = A\mathbf{u}_2 = \begin{bmatrix} \frac{x_3(t)}{C_1} \\ \frac{-x_3(t)}{C_2} \\ \frac{x_2(t)-x_1(t)-Rx_3(t)}{L} \end{bmatrix} \tag{3.49}$$

$$A_{u_3} = A\mathbf{u}_3 = \begin{bmatrix} \frac{x_3(t)}{C_1} \\ 0 \\ \frac{E-x_1(t)-Rx_3(t)}{L} \end{bmatrix} \tag{3.50}$$

$$A_{u_4} = A\mathbf{u}_4 = \begin{bmatrix} \frac{-x_3(t)}{C_1} \\ 0 \\ \frac{x_1(t)-Rx_3(t)}{L} \end{bmatrix} \tag{3.51}$$

$$A_{u_5} = A\mathbf{u}_5 = \begin{bmatrix} \frac{-x_3(t)}{C_1} \\ \frac{x_3(t)}{C_2} \\ \frac{E-x_1(t)-x_2(t)-Rx_3(t)}{L} \end{bmatrix} \tag{3.52}$$

$$A_{u_6} = A\mathbf{u}_6 = \begin{bmatrix} 0 \\ \frac{-x_3(t)}{C_2} \\ \frac{x_2(t) - Rx_3(t)}{L} \end{bmatrix} \quad (3.53)$$

$$A_{u_7} = A\mathbf{u}_7 = \begin{bmatrix} 0 \\ 0 \\ \frac{E - Rx_3(t)}{L} \end{bmatrix} \quad (3.54)$$

$$\dot{X} = A_{u_0}X + B_0u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \quad (3.55)$$

$$\dot{X} = A_{u_1}X + B_1u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{1}{C_2} \\ 0 & -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \quad (3.56)$$

$$\dot{X} = A_{u_2}X + B_2u = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & -\frac{1}{C_2} \\ -\frac{1}{L} & \frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \quad (3.57)$$

$$\dot{X} = A_{u_3}X + B_3u = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & \frac{1}{C_2} \\ -\frac{1}{L} & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \quad (3.58)$$

$$\dot{X} = A_{u_4}X + B_4u = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & 0 \\ -\frac{1}{L} & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \quad (3.59)$$

$$\dot{X} = A_{u_5}X + B_5u = \begin{bmatrix} 0 & 0 & -\frac{1}{C_1} \\ 0 & 0 & \frac{1}{C_2} \\ \frac{1}{L} & -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \quad (3.60)$$

$$\dot{X} = A_{u_6}X + B_6u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{1}{C_2} \\ 0 & -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \quad (3.61)$$

$$\dot{X} = A_{u_7}X + B_7u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \quad (3.62)$$

4 Conclusão

Concluimos que esse eh o capitulo 4...

colocar um plano de trabalho e cronograma

possibilidade de uma aplicacao do trabalho usando um conversor de tensao

	2022				2023			
	1	2	1	2	1	2		
extensão de caso tempo final								
elaboração de um artigo com resultados								
aplicação com maior número de estados								
experimental								
elaboração de tese final(defesa)								

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Appendix A - Tópicos de Dilema Linear

A.1 Modelo de simulação

```
+engine
├─ CORES.m
├─ fun_custo_patino.m
├─ get_config_sim_matheus.m
├─ get_config_sim_patino_1.m
├─ get_config_sim_patino_2.m
├─ get_config.m
├─ get_otmin_opt.m
├─ get_ts.m
├─ get_x0.m
├─ get_xr.m
├─ nonlincon_patino.m
├─ otmin.m
├─ sim_1.m
├─ sim_n.m
└─ +mpc
    ├─ construcao_modelo_instantes.m
    ├─ matrizes_ss_mpc_dualmode_switching.m
    └─ mpc_dualmode_switching.m
```

A.1.1 get_x0

funcao de calculo da condicao inicial

A.1.2 get_xf

Texto do modelo de simulação sendo colocado aqui.

Annex A - Exemplo de um Primeiro Anexo

A.1 Uma Seção do Primeiro Anexo

Algum texto na primeira seção do primeiro anexo.

FOLHA DE REGISTRO DO DOCUMENTO

1. CLASSIFICAÇÃO/TIPO TD	2. DATA 25 de março de 2015	3. DOCUMENTO Nº DCTA/ITA/DM-018/2015	4. Nº DE PÁGINAS 39
5. TÍTULO E SUBTÍTULO: Robust control of linear systems with Switched Actuators Subjected to Dwell-Time Constraints			
6. AUTOR(ES): Daniel Vieira			
7. INSTITUIÇÃO(ÕES)/ÓRGÃO(S) INTERNO(S)/DIVISÃO(ÕES): Instituto Tecnológico de Aeronáutica – ITA			
8. PALAVRAS-CHAVE SUGERIDAS PELO AUTOR: Cupim; Cimento; Estruturas			
9. PALAVRAS-CHAVE RESULTANTES DE INDEXAÇÃO: Cupim; Dilema; Construção			
10. APRESENTAÇÃO: (X) Nacional () Internacional ITA, São José dos Campos. Curso de Mestrado. Programa de Pós-Graduação em Engenharia Aeronáutica e Mecânica. Área de Sistemas Aeroespaciais e Mecatrônica. Orientador: Prof. Dr. Adalberto Santos Dupont. Coorientadora: Prof ^a . Dr ^a . Doralice Serra. Defesa em 05/03/2015. Publicada em 25/03/2015.			
11. RESUMO: Atuadores chaveados são difíceis de lidar por conta de algumas restrições características de sua natureza, porém eles são eficientes em inúmeras aplicações da engenharia, sendo aplicado, por exemplo, em sistemas de controle de atitude de veículos espaciais. Assim, saber utilizar desse tipo de atuador se faz importante e necessário. O principal problema a ser resolvido nessa tese é o projeto de um controlador que garanta a robustez de sistema com atuadores chaveados submetidos a restrições temporais que limitam a janela de resposta de seus comandos. Nessa abordagem, é feita uma busca de trajetória alvo de ciclo limite e, então, é calculado o erro dos estados e aplicada a técnica de controle preditivo para correção do erro e levar o systema à trajetória cíclica alvo.			
12. GRAU DE SIGILO: (X) OSTENSIVO () RESERVADO () SECRETO			